

h1. (a) $S_1 (0, 0, 1, 0, 0, 0, 1)$ $e -$

$S_2 (1, 1, 0, 0, 0, 0)$

$S_3 (0, 0, 0, 1, 1, 0)$

~~S_4~~ $(1, 0, 1, 0, 1, 0)$

$q (1, 0, 1, 0, 1, 0)$

(b) II)		S_1	S_2	S_3	$q.$	$h_1(x)$	$h_2(x)$	$h_3(x)$
0	a	0	1	0	1	1	2	2
1	b	0	1	0	0	3	5	1
2	c	1	0	0	1	5	2	0
3	d	0	0	1	0	1	5	5
4	e	0	0	1	1	3	2	4
5	f	1	0	0	0	5	5	3

(c). $h(3)$

(d) Signature: use $h_3(x)$

	s_1	s_2	s_3	q
$h_1(x)$	5	1	1	1
$h_2(x)$	2	2	2	2
$h_3(x)$	0	1	4	0

estimate

$$J(q, s_1) = \frac{2}{3} = \frac{1}{3}$$

1.2.

对, 对.

$$(a). FN = 1 - Pr(s)$$

对, 错

$$FN(r=3, b=10) = 1 - 0.98 \approx 0.02 \approx 2\%$$

$$r=b, b=20, \quad = 1 - 0.918 \approx 7.2\%$$

$$r=5, b=50 \quad = 1 - 0.999 \approx 0.01\%.$$

$$r=3, b=10 \text{ and } r=5, b=50.$$

$$(b) \quad Fp = P(s).$$

错 $\rightarrow$ 对.

$$r=3, b=10$$

$$23\% \quad X$$

$$b=20$$

$$1\% \quad \checkmark$$

$$b=50$$

$$11\% \quad \checkmark$$

1.3 increase p_1 to 1

decrease p_2 to 0

$$P_{fp} = C(1 - e^{-k \frac{n}{m}})^k$$

$$m \uparrow$$

k

Misra - Gries.

Algorithm

(a).

45	13	14	7
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counter 1 1 1

45, 13, 14, 45, 7.

counter

45	6		
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2 1

10. 45, 6

(b), set theoretical number of decrease
step n

total number remain is n'

k counters.

total number of stream is m .

$$m = n' + (k+1) \cdot n,$$

$$n = \frac{m - n'}{k+1}$$

$$n = \frac{8 - 3}{5} = \frac{5}{5} \rightarrow n = 1$$

2.3.

(a).

	45	13,	14	7	10,	b.
$h_1(x)$	0	1	2	1	1	0
$h_2(x)$	1	0	2	0	0	1
$h_3(x)$	1	1	1	1	1	1
$s_1(x)$	1	-1	-1	-1	-1	1
$s_2(x)$	1	1	1	1	1	1
$s_3(x)$	-1	-1	1	-1	-1	-1

(b) counter matrix

0,	1,	2,
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$h_1(x)$	1+1	-1-1-1	-1
$h_2(x)$	1+1+1	1+1	1
$h_3(x)$		-1-1+1	
		-1-1	

	0	1	2
$h_1(x)$	2	-3	-1
$h_2(x)$	3	2	1
$h_3(x)$	0	-4	0

45: $h(45) = (0, 1, 1),$
 $(2, 1, -2),$

$$2 \times 1 = 2$$

$$1 \times 1 = 1$$

$$-1 \times -2 = 2$$

$$\text{med}(2, 1, 2)$$

$$= 2$$



$$h(13) = (1, 0, 1)$$

$$\text{med}(-1 \times (-3), 1 \times (3), -1 \times (-4))$$

$$3, 3, 4.$$

$$(3).$$

web graph.

$$(a) \quad A = \begin{bmatrix} \frac{1}{2} & 0 & 0 & 0 \\ \frac{1}{2} & 1 & \frac{1}{2} & \frac{1}{2} \\ 0 & 0 & \frac{1}{2} & 0 \\ 0 & 0 & 0 & \frac{1}{2} \end{bmatrix}$$

$$(b) \quad \begin{bmatrix} \frac{1}{2} & 0 & 0 & 0 \\ \frac{1}{2} & 1 & \frac{1}{2} & \frac{1}{2} \\ 0 & 0 & \frac{1}{2} & 0 \\ 0 & 0 & 0 & \frac{1}{2} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix}$$

$$\frac{1}{2} x_1 = x_1$$

$$\frac{1}{2} x_1 + x_2 + \frac{1}{2} x_3 + \frac{1}{2} x_4 = x_2$$

$$\frac{1}{2} x_3 = x_3 \quad \geq 0$$

$$\frac{1}{2}xy = xy \quad \geq 0$$

$$x_1 = 0, x_2 = 1, x_3 \geq 0, xy = 0$$

$$x_2 = x_2 \geq 0$$

$$x_1 + x_2 + x_3 + xy = 1$$

$$r_0 = \begin{bmatrix} 0 \\ 1 \\ 0 \\ 0 \end{bmatrix}$$

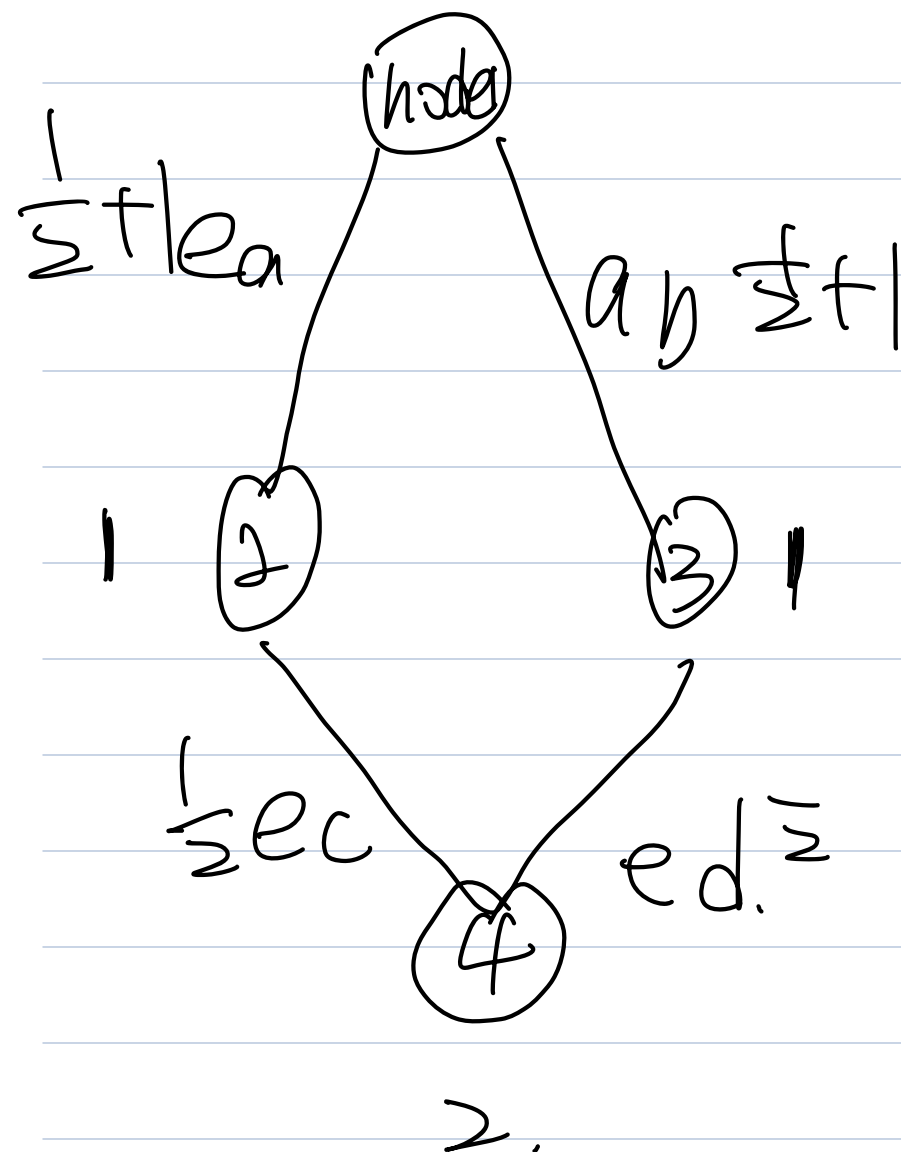
using last time result

as r , multiply

$$r(i+1) = M r(i)$$

Threshold

community detection.



$$e_b = e_a = 1.5$$

$$e_c = e_d = 0.5$$

(b)

$$L = D - M.$$

$P(x)$ node

$1 \rightarrow 2$	$1 \rightarrow 3$	0.4	3
$1 \nrightarrow 2$	$1 \nrightarrow 3$	0.1	1
$1 \rightarrow 2$	$1 \nrightarrow 3$	0.4	2
$1 \nrightarrow 2$	$1 \rightarrow 3$	0.1	2

$$3 \times 0.4 + 1 \times 0.1 + 2 \times 0.4 + 2 \times 0.1$$

1.2

0.1

0.8

0.2

$$= 2.3$$

	P_1	P_2	P_3	P_4
V_1	5	0	2	0
ΔV_2	3	4	0	0
	1	0	1	1

v_3	4	0	1	4
v_4	1	3	0	0
v_5	5	4	0	0

0x0

$$\text{Cosine Similarity}(v_1, v_2) = \frac{5 \times 3 + 0 \times 4 + 2 \times 0}{\sqrt{29} \cdot \sqrt{25}}$$

$$= \frac{15}{5\sqrt{29}} = \frac{1}{\sqrt{29}}$$

$$= \frac{\sqrt{3}}{3} \approx 0.577$$

$$\text{Cosine Similarity}(v_2, v_3) = \frac{12}{\sqrt{25} \cdot \sqrt{33}}$$

$$= \frac{12}{5\sqrt{33}}$$

$$\approx 0.4177$$

$$> 12$$

$$V_2, V_4 = \frac{31}{5 \cdot \sqrt{10}}$$

$$= \frac{3}{\sqrt{10}} = 0.948$$

$$V_2, V_5 = \frac{15+16}{5\sqrt{41}}$$

$$= \frac{31}{5\sqrt{41}} = 1.21,$$

V_5 .

(b)

a_1

31b

$$g_b = \frac{11}{11}$$

$$= 3.27$$

$$V_2(b_{u2}) = \frac{3+4}{2} - g_b$$

$$= 3.5 - 3.27$$

$$= 0.23$$

$$b(p_1^{iter}) = \frac{18}{5} - g_b$$

$$= 0.33$$

c)

$$\begin{aligned}
 RMES &= \sqrt{\frac{(1-4)^2 + (5-3)^2 + (5-4)^2 + (3-2)^2 + (4-5)^2}{5}} \\
 &= \sqrt{\frac{9 + 4 + 1 + 1 + 1}{5}} \\
 &= \sqrt{3.2}
 \end{aligned}$$

4.2. $e_{ij} = r_{ij} - q_i p_j$

$$q_i \leftarrow q_i + \eta (e_i r_{ij} - \lambda_1 q_i)$$

$$p_j' \leftarrow p_j + \eta (e_{ij} q_i' - \lambda_2 p_j)$$

$$q_i' \leftarrow q_i + \eta (e_{ij} p_j - \lambda_1 q_i)$$

2020

bit vector.

1.1 S1 (1, 1, 1, 1, 1)
 S2 (1, 1, 1, 1, 0)
 S3 (1, 1, 0, 1, 0)
 q (0, 1, 0, 1, 0)

count-vector.

0 1 2 3 4.
 S1 (1, 1, 1, 1, 1)
 S2 (1, 3, 1, 1, 0)
 S3 (1, 2, 0, 1, 0)
 q (0, 3, 0, 1, 0)

1.2.	ID	S1	S2	S3	q	$h_1(x)$	$h_2(x)$	$h_3(x)$
0	0	1	1	1	0	2	1	3
1	1	1	1	1	1	0	2	0
2	2	1	1	0	0	3	3	2
3	3	1	1	1	1	1	4	4

4 4 1 0 0 0 4 0 1

Signature	s_1	s_2	s_3	q
$h_1(x)$	0	0	0	0
$h_2(x)$	0	1	1	2
$h_3(x)$	0	0	0	0

Sim hash.

count-vector.

	0	1	2	3	4
s_1	1	1	1	1	1
s_2	1	3	1	1	0
s_3	1	2	0	1	0
q	0	3	0	1	0

$$r_1 (1, -1, 0, 1, -1)$$

$$r_2 (0, 0.5, 0, -1, 0.5)$$

$$r_3 (0, -0.5, 1, 0, 1)$$

$$r \cdot s \begin{cases} 1 & \cos \theta \geq 0 \\ 0 & \cos \theta < 0 \end{cases}$$

Signature.

	r_1	r_2	r_3		r_1	r_2	r_3
s_1	0	0	1.5	s_1	1	1	1
s_2	-1	0.5	-0.5	s_2	0	1	0
s_3	0	0	-1	s_3	1	1	0
q	-2	0.5	-1.5	q	0	1	0

3.

$$1 - p(s) < 0.15$$